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Dynamic optimization of e-commerce supply chains for fresh products with blockchain and reference effect

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ABSTRACT

Due to information asymmetry, consumers often rely on reference freshness when purchasing fresh products online. We employ blockchain technology to address this issue and enhance transparency in the fresh product supply chain. Specifically, using product freshness and goodwill as state variables, we develop a fresh product supply chain model involving a supplier and an e-commerce platform, based on differential game theory. By comparing traditional supply chain models with those utilizing blockchain technology, this study reveals that blockchain implementation encourages suppliers to increase their freshness-keeping efforts during distribution. Meanwhile, the e-commerce platform may reduce its advertising efforts. This strategic shift can result in a higher long-term equilibrium freshness compared to scenarios without blockchain implementation. Additionally, the research demonstrates that goodwill significantly improves when blockchain investment efficiency reaches a sufficient threshold. However, the study also highlights how reference freshness can lower the level of blockchain traceability. These findings provide valuable insights into the application of blockchain in managing fresh products on e-commerce platforms.

1. Introduction

Freshness is an essential indicator for fresh products (Cai et al., 2010; Antony et al., 2018). Unlike offline retail, selling fresh products to consumers online requires cold chain logistics for storage and transportation to maintain freshness (He et al., 2019). Progressive investment in cold chain logistics ensures better product freshness, encouraging more consumers to purchase fresh products through e-commerce platforms. It also drives enterprises to enhance their online channels for sales promotion and competitive advantage (Wang et al., 2017). According to the DataIntelo Report, the global fresh food e-commerce market size was valued at roughly USD 75 billion in 2023. It is expected to reach USD 250 billion by 2032, growing at a compound annual growth rate of 14% during the forecast period.

Consumers typically assess freshness through the appearance, color, and taste of products such as fruits and vegetables. However, in online shopping, direct evaluation is not possible. Consumers rely on online reviews, product images, and videos to form expectations about freshness. Upon receiving the product, they compare its freshness to their expectations, a process commonly referred to as the reference freshness effect (RFE). Satisfaction increases when the product's freshness exceeds expectations, enhancing goodwill. Over time, goodwill may naturally decline due to consumer forgetfulness and exposure to alternative advertised or promoted products (Zhao et al., 2023).

Goodwill plays a vital role in e-commerce, as consumers often rely on a platform or brand's goodwill to assess product freshness (Liu et al., 2021a; Li et al., 2023b). E-commerce platforms typically invest in advertising to build and maintain goodwill, influencing consumer purchasing decisions (Sigue and Gromova, 2023; Li et al., 2023a; Hemmati et al., 2024). For instance, Amazon employs a variety of advertising formats — such as Amazon Live, video, audio, and display ads — to deliver clear product details, helping consumers set accurate expectations for product freshness (Chenavaz and Jasimuddin, 2017). These advertising efforts clarify product features and enhance brand awareness and market presence (Xu et al., 2023; Khorshidvand et al., 2021).

Despite these efforts, fresh e-commerce platforms face challenges with discrepancies between advertised and actual product freshness, which directly impact consumer trust and purchase decisions. A 2021 Consumer Reports survey highlighted that 40% of U.S. consumers expressed dissatisfaction with the freshness and consistency of delivered products, mainly fruits and vegetables. Many platforms use QR codes

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to mitigate consumer concerns. However, these codes can be easily replicated, rendering commodity traceability ineffective (Zhao and Li, 2023; Hou et al., 2024). For example, in 2019, Whole Foods, a leading U.S. supermarket chain, faced allegations of selling meat products with altered expiration dates. Offering authentic traceability information — such as production details, storage data, and transportation records — on fresh products can help consumers assess product quality.

To address these issues, some fresh product supply chain (FPSC) firms have begun implementing blockchain technology (BT) for product traceability. Blockchain enables platforms to monitor the production and supply flow of fresh products, from farm to store shelf. It tracks the origin, handling, storage, transportation, and arrival times of products (Shen et al., 2020). As an emerging traceability technology, blockchain offers decentralization, transparency, encryption, and tamper resistance (Kamble et al., 2019). Platforms like GrubMarket have implemented comprehensive digital upgrades, realizing a "farm-to-table" business model that positions them as leaders in the U.S. fresh product retail market.

The higher level of traceability enhances consumers' ability to assess product quality, increasing their willingness to pay for reliable information. However, excessive information can cause decision fatigue, which undermines consumers' ability to make informed choices. In addition, despite BT providing robust and transparent tracking functionality, it requires a significant investment (Chang et al., 2018). Thus, FPSC members must balance and optimize traceability levels. To address this challenge, we apply differential game models to the FPSC, comprising a supplier and an e-commerce platform. The traditional FPSC is model N, while the blockchain-based FPSC is model B. Considering the consumer's RFE and utilizing state equations to characterize product freshness and goodwill, our work attempts to answer the following questions:

(i) RQ1: What are the optimal dynamic trajectory rules for product freshness and goodwill under different scenarios?

(ii) RQ2: How does RFE affect the optimal decisions of each member and the implementation of BT in the FPSC?

(iii) RQ3: How does adopting BT affect the optimal choices of chain members and the steady-state value of freshness and goodwill?

First, we theoretically derive the equilibrium strategies and profits in both models. Next, we compare these equilibrium strategies between the two models. Furthermore, we identify the role of RFE and BT in the FPSC, leading to noteworthy findings: (i) As RFE increases, the optimal freshness-keeping effort level increases while advertising effort decreases, and the platform may reduce traceability of BT; (ii) The platform must carefully evaluate the efficiency of its investment, as BT benefits FPSC members only when the efficiency reaches a high threshold; (iii) Blockchain implementation increases freshness-keeping efforts while typically decreasing advertising efforts; (iv) Under steady-state conditions, the level of freshness achieved with blockchain exceeds that of the traditional model. However, goodwill will increase following the implementation of BT only if its investment efficiency is sufficiently high.

The remainder of this paper is structured as follows: Section 2 presents a review of related literature. Section 3 outlines the methodology and model settings. Section 4 provides equilibrium analyses for two cases. Section 5 compares the blockchain model with the baseline model. Section 6 performs the numerical analysis. Finally, Section 7 offers managerial insights and future research directions.

2. Literature review

Three interrelated domains of literature relevant to our study are consumer reference effect behaviors, advertising investment and its relationship with goodwill, and blockchain implementation in supply chains.

2.1. Consumer reference effect behavior

Understanding consumer behavior is crucial for optimizing a firm's profitability. As behavioral economics suggests, consumers evaluate gains and losses relative to a relevant reference point when making purchasing decisions (Yiran and Jiajia, 2021). In studies of consumer purchasing behavior, He et al. (2016) found that the reference effect significantly influences consumer decisions. The reference effect highlights consumers' heightened sensitivity to specific product attributes, such as price or quality. Pre-purchase expectations of these attributes significantly influence consumer decisions and shape companies' strategic choices.

The reference price effect has been studied extensively. Dye and Yang (2016) and Li and Teng (2018) investigated dynamic pricing under the influence of reference prices. Martín-Herrán et al. (2012) explored the impact on myopic retailers (i.e., those who ignore the price reference effect in consumer purchasing behavior) when consumers exhibit a price reference effect. Zhang et al. (2013) examined how advertising influences reference prices. They proposed a second-tier dynamic cooperative advertising model to evaluate its impact on the operational decisions of supply chain members.

Research on the reference quality effect remains limited, as most current studies are based on experimental and empirical methods (Han et al., 2022). Chenavaz and Jasimuddin (2017) investigated the influence of quality reference effect on the firms' quality management. Liu et al. (2016) examined the impact of reference quality on manufacturer and retailer's product quality and pricing strategies. Xue et al. (2017) suggested that continuously increasing quality investment throughout the planning cycle is not optimal for firms. They argued that firms should instead focus on consumers' reference-quality effects concerning dynamic pricing and product quality in the context of inflation. In their research, He et al. (2018) applied the consumers' reference-quality impact to the production decision of Online to Offline (O2O) enterprises. However, the psychological impact of consumer reference quality remains undervalued in the FPSC despite its importance, with only a few studies directly relevant to our research. Li and Hou (2019) studied dynamic quality investment and inventory replenishment for perishable goods, considering the reference quality effect in the O2O environment. Zhao et al. (2023) analyzed the optimal selling strategies considering the consumer reference freshness effect within a fresh food supply chain.

The difference between actual and reference freshness plays a pivotal role in a consumer's subsequent willingness to purchase more of the product. When consumers shop online, the reference freshness, often shaped by the brand's goodwill, holds considerable sway over buyer behavior. However, the existing literature has not explored in depth the dual impact of RFE and BT on product demand. This gap in the literature limits our understanding of the complex relationship between BT implementation and consumer reference behavior.

2.2. Advertising investment and goodwill

As fierce competition in the market intensifies, more businesses are using advertising to attract consumers (Xu et al., 2021). Therefore, numerous scholars focus on advertising equilibrium strategies under different conditions, such as considering advertising delay effort, platform goodwill, and brand goodwill (Wu et al., 2018; Wu, 2019). Using the Nerlove–Arrow model, Machowska (2019) analyzed the immediate and delayed impacts of cooperative advertising strategies on the evolution of goodwill in a competitive duopoly market. Wu et al. (2020) examined advertising investment decisions for the consignment platform, considering delayed advertising effects on the dynamics of platform goodwill.

Some critical assumptions of the goodwill model are highly relevant to our research. According to Liu et al. (2021a), the demand for fresh products is often influenced by the goodwill of the retailer's platform,

as customers are more inclined to place orders on platforms with higher goodwill. Dye (2020) included the effects of pricing, psychological inventory, product freshness index, and advertising goodwill into a dynamic decision-making framework for perishable products over an unlimited time horizon. They emphasized that goodwill is influenced by a company's current and past advertising efforts.

Consumers face challenges in assessing the freshness of food when shopping online. They rely on platform goodwill and form psychological expectations regarding product freshness. Zhao et al. (2023) suggested that reference freshness significantly influences a retailer's goodwill. Therefore, we leverage the characteristics of goodwill while building upon previous studies by El Ouardighi (2014) and Zhao et al. (2023). In this context, we include BT traceability in the goodwill function. Adopting BT reduces freshness uncertainty in entirely online sales, as highlighted by De Giovanni (2020). This further reduces advertising efforts, reflecting the impact of the BT implementation on the trajectory of product goodwill.

2.3. Blockchain implementation in supply chains

With the development of BT, a substantial body of literature has examined the role of this technology in supply chain management from multiple perspectives. Traceability is a primary motivation for applying BT in supply chains (Kamble et al., 2020). Numerous studies have developed models for incorporating BT into supply chains. According to Hastig and Sodhi (2020), blockchain facilitates information transparency and product traceability in the supply chain. Wu and Yu (2023) examined the effect of BT on platform supply chains, focusing on information transparency and transaction costs. Tian and Hu (2023) investigated the influence of BT on pricing and freshness investment decisions in online game supply chains, particularly in scenarios with information asymmetry. Considering BT's potential to eliminate freshness information asymmetry, Zhang et al. (2024) adopted a game theory model to explore manufacturers' optimal information strategies in a dual-channel supply chain.

In the FPSC field, He et al. (2024) examined how the implementation of BT influences pricing and profitability for participants in the global fresh produce market. Liu et al. (2021b) investigated the resale and consignment decisions of fresh products and analyzed the competition between traditional and e-commerce channels, employing BT to track the reputation and freshness of fresh products. Wu et al. (2023) analyzed the strategies of suppliers, third-party logistics service providers, and e-tailers joining the BT platform in the fresh agricultural product supply chain. Xu et al. (2024) investigated the relationship between channel encroachment decisions and the tactics used by the suppliers of fresh agricultural products to join the BT platform. Tan et al. (2023) examined the implementation strategies of fresh agricultural product suppliers on the BT platform, taking into account factors such as BT cost and perceived fresh product quality.

The studies cited above were primarily conducted in static analytical environments. In contrast, our research considers the dynamics of fresh product freshness changes and goodwill trajectory, enabling decisions to adapt to evolving conditions. From the perspective of dynamic analytical contexts, Liu et al. (2021a) employed BT to monitor the goodwill and freshness of fresh products as well as their resale and consignment choices in a traditional competitive context. Li et al. (2023b) demonstrated the importance of adopting BT to improve the freshness and reputation of fresh products from a dynamic perspective. While Li et al. (2023b) focused on improving freshness and reputation dynamically, our study emphasizes that implementing BT can significantly reduce product freshness uncertainty for products sold purely online, thereby affecting consumers' purchasing decisions. Moreover, incorporating the traceability of BT into the goodwill function ensures that traceability-related decision-making processes are dynamic.

This study makes several contributions to the existing body of research. Firstly, we explore the combination of RFE and BT and analyze how their combined effects affect the decisions of supply chain members. Secondly, we creatively incorporate the traceability of BT implementation into the goodwill accumulation function, providing a new perspective for understanding the relationship between freshness, goodwill, and BT implementation. We propose an innovative method to evaluate changes in demand functions before and after BT implementation. By conducting this analysis, we aim to accelerate the application and integration of BT in the fresh product e-commerce market. Finally, unlike previous studies that mainly focus on static decision-making environments, we develop a dynamic decision-making model to analyze BT implementation in the online sales environment of fresh products and study its long-term impact on product freshness and goodwill trajectories. These results provide valuable references and guidance for the development of blockchain businesses.

3. Model description

We study an FPSC comprising a fresh product supplier ('supplier', subscript s) and an E-commerce platform ('platform', subscript r). The supplier sells its items to customers via the online channel. The two parties engage in a revenue-sharing contract. The platform charges the supplier a fixed commission fee and takes responsibility for advertising expenses, while the supplier maintains product freshness. Typically, the commission fee and sales price are pre-determined (ex-ante) and treated as exogenous variables in this model (Xu et al., 2021). Fig. 1 depicts the FPSC's decision-making process.

3.1. Dynamic freshness

Consumers generally prefer products with high freshness, which motivates suppliers to actively invest in freshness-keeping efforts. For instance, research (Liu et al., 2021a; Luo et al., 2023) highlights advancements in freshness-keeping technology. Continuous advancements in freshness-keeping technology contribute to maintaining product quality. Therefore, the freshness of these products can be effectively described dynamically. Nonetheless, it is essential to recognize that perishable goods inherently deteriorate over time. This natural process results in reduced freshness. The dynamic process of fresh product freshness can be formulated as follows:

$$\dot{Q}(t) = \alpha\mu(t) - \eta Q(t), Q(0) = Q_0, \tag{1}$$

where $Q(t)$ denotes the product freshness at time t, and Q_0 is the initial level of freshness. $\mu(t)$ represents freshness-keeping efforts at time t, $\alpha > 0$ signifies the impact of freshness-keeping efforts on product freshness, and $\eta > 0$ denotes the decay rate. Moreover, as freshness-keeping efforts $\mu(t)$ increase, the process becomes more complex, leading to higher marginal costs. Therefore, the freshness-keeping cost at time t is represented by the following formula:

$$C(\mu(t)) = \frac{k_f \mu^2(t)}{2}, \tag{2}$$

where $k_f > 0$ represents the cost coefficient associated with freshness-keeping effort (Song et al., 2021).

3.2. Dynamic goodwill

Consumers often rely on reference freshness value $r(t)$ when shopping online. Existing literature, such as Li and Hou (2019), demonstrates the relationship between reference freshness and goodwill as follows:

$$r(t) = \xi G(t), \tag{3}$$

where $\xi > 0$ is the influence coefficient of goodwill on reference freshness.

The RFE, that is, $Q(t) - r(t)$, occurs when consumers compare the product's freshness to their reference freshness upon receiving it. We use a Nerlove–Arrow model (Zhao et al., 2023) to represent the

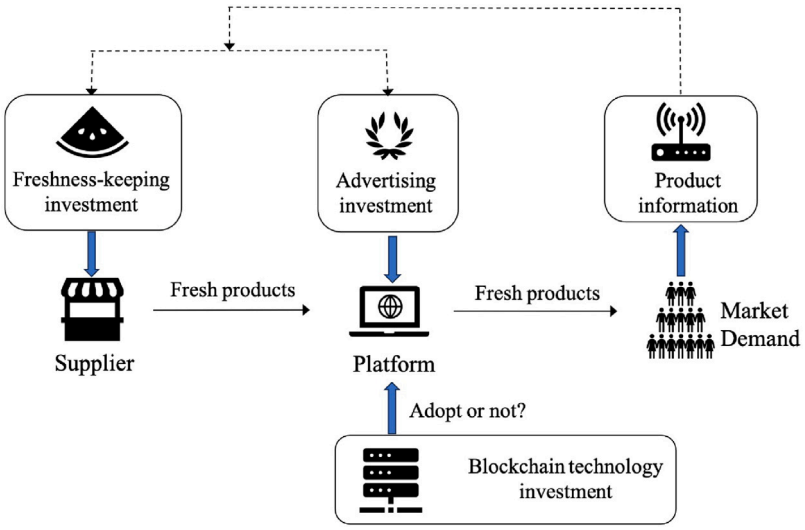


Fig. 1. The specific decision-making process of the FPSC.

goodwill dynamics of the traditional model, as shown below:

$$\dot{G}^N(t) = \beta a(t) + \delta(Q(t) - r(t)) - \varphi G(t), G(0) = G_0, \tag{4}$$

where $G(t)$ is the accumulated goodwill over time t, $a(t)$ is the advertising effort level at time t, $\beta > 0$ is the influence coefficient of the advertising effort, $\varphi > 0$ is the natural decline rate of goodwill, G_0 is the initial goodwill, and $\delta > 0$ is the freshness efficiency to goodwill accumulation. Specifically, the term $\delta(Q(t) - r(t))$ indicates that product freshness exceeding consumer expectations contributes positively to goodwill, while freshness below expectations has a negative impact. Next, we define $\sigma = \delta\xi + \varphi$, and rewrite the $G^N(t)$ with $r(t)$ as:

$$\dot{G}^N(t) = \beta a(t) + \delta Q(t) - \sigma G(t), G(0) = G_0. \tag{5}$$

BT offers unique advantages to the FPSC, particularly in terms of traceability and marketing. Blockchain traceability builds increased consumer trust in the freshness of product information, thus boosting their confidence in making purchases. From a marketing standpoint, traceability within the supply chain ensures consistent product delivery to customers and guarantees product safety. Thus, blockchain traceability strengthens consumer loyalty and enhances platform goodwill. According to the previous research with the Nerlove–Arrow model (Liu et al., 2021b), blockchain-enabled traceability goodwill can be characterized as follows:

$$\dot{G}^B(t) = \beta a(t) + \delta Q(t) + f b(t) - \sigma G(t), G(0) = G_0, \tag{6}$$

where $b(t)$ represents the traceability level of BT at time t, and $f > 0$ is the coefficient influencing the traceability level of the goodwill.

As the traceability level increases, so does the marginal cost of verifying information, as noted by Wu et al. (2023), Li et al. (2023b), the higher the traceability level, so does the marginal cost of verifying information. As a result, the cost associated with adopting BT at time t can be expressed as:

$$C(b(t)) = \frac{k_b b^2(t)}{2}, \tag{7}$$

where $k_b > 0$ is the cost coefficient of the traceability level.

3.3. Customer demand function

Consumers prefer purchasing products with superior freshness levels and are inclined to choose a more reputable sales platform for their shopping requirements. Market demand was then the same as previously (Li et al., 2023b; Liu et al., 2021a) used literature incorporating freshness-related factors while excluding price effect. In the non-blockchain-based strategy (model N), consumers have a prior belief of freshness based on the goodwill of the platform as they cannot observe the product's freshness directly. Hence, we can formulate the market demand function $D^N(t)$ as follows:

$$D^N(t) = A + \omega G(t), \tag{8}$$

where A represents the initial market size, and ω indicates the sensitivity of the demand to goodwill. Under Strategy B, the platform utilizes BT to provide consumers with access to freshness information, ensuring they can verify product quality before making a purchase (Choi et al., 2020). Hence, we have

$$D^B(t) = A + \theta(Q(t) - r(t)) + \omega G(t), \tag{9}$$

where $\theta > 0$ reflects the sensitivity of the demand to RFE.

The state variables, decision variables, and relative parameters adopted in the models are summarized in Table 1.

4. Model solving

In this section, we first establish a baseline model for traditional FPSC operations. The initial model development identifies optimal freshness-keeping and advertising effort decisions. Subsequently, we analyze freshness disclosure strategy models under blockchain adoption, focusing on equilibrium freshness-keeping effort, advertising effort, and traceability levels.

4.1. Strategy analysis for traditional case (Model N)

For a fresh FPSC with an infinite time horizon, we consider a positive discount rate, ρ. In this case, the supplier initially determines the freshness-keeping effort, represented as $\mu^N(t)$, and the platform subsequently allocates the advertising effort, denoted as $a^N(t)$. Finally, the quantities of products consumers purchase are denoted as $D^N(t)$. The objective functions of the supplier and the platform are given as follows:

$$V_s^N(t) = \int_0^\infty e^{-\rho t} \left\{ \pi_s D^N(t) - \frac{k_f \mu^2(t)}{2} \right\} dt \tag{10}$$

$$V_r^N(t) = \int_0^\infty e^{-\rho t} \left\{ \pi_r D^N(t) - \frac{k_m a^2(t)}{2} \right\} dt \tag{11}$$

Table 1
Parameters and descriptions.

Notation	Description
Decision variables	
$\mu^j(t)$	Freshness-keeping efforts of the supplier for the case j at time t
$a^j(t)$	Advertising efforts of the platform for the case j at time t
$b(t)$	Traceability level of the platform at time t
Parameters	
π_s	Marginal benefit of the supplier, $\pi_s > 0$
π_r	Marginal benefit of the platform, $\pi_m > 0$
Q_0, G_0	Initial freshness and goodwill
$r(t)$	Consumers' reference freshness at time t
A	Basic market size
θ, ω	Influence coefficient of freshness and goodwill on demand, $\theta, \omega > 0$
α	Influence coefficient of the freshness-keeping effort on freshness, $\alpha > 0$
β	Influence coefficient of the advertising effort on goodwill, $\beta > 0$
f	Influence coefficient of the traceability level on goodwill, $f > 0$
ρ	Discount rate, $\rho > 0$
η, φ	Decay rate of the product freshness and goodwill
δ	Freshness efficiency to goodwill accumulation, $\delta > 0$
ξ	Goodwill efficiency to reference freshness, $\xi > 0$
σ	Positive parameter, $\sigma > 0$
k_f, k_m, k_b	Cost coefficient of the freshness-keeping, advertising and traceability investment
Objective functions	
V_s^j	Profit function of the supplier for the case j
V_r^j	Profit function of the platform for the case j
State variables	
$Q^j(t)$	Accumulated freshness for the case j at time t
$G^j(t)$	Accumulated goodwill for the case j at time t
$D^j(t)$	Market demand function for the case j at time t
Model index	
t	Selling time
j	$i \in \{N, B\}$, where $j = N$ for traditional model, $j = B$ for blockchain model

where π_s is the marginal benefit of the supplier, and π_r is the marginal benefit of the retailer. Other variables are defined as previously described.

By combining the dynamics described in Eqs. (1)–(5), (8), (10) and (11), we derive the following decision problems for the supplier and the platform under Model N.

$$\max_{\mu^N(t)} \int_0^\infty e^{-\rho t} \left\{ \pi_s(A + \omega G(t)) - \frac{k_f \mu^2(t)}{2} \right\} dt \tag{12}$$

$$\max_{a^N(t)} \int_0^\infty e^{-\rho t} \left\{ \pi_r(A + \omega G(t)) - \frac{k_m a^2(t)}{2} \right\} dt \tag{13}$$

Using the Pontryagin maximization principle, we solve Eqs. (12)–(13) to derive Theorem 1, which yields the optimal freshness-keeping and advertising efforts.

Theorem 1. *In the traditional FPSC, the following equilibrium solutions hold:*

(i) The optimal strategies for the supplier and the platform are as follows: $\mu^{N*} = \frac{\alpha\delta\omega\pi_s}{(\rho+\sigma)(\eta+\rho)k_f}$, $a^{N*} = \frac{\beta\omega\pi_r}{(\sigma+\rho)k_m}$.

(ii) The time trajectories of product freshness and goodwill are as follows: $Q^{N*}(t) = (Q_0 - Q_\infty^N)e^{-\eta t} + Q_\infty^N$, $G^{N*}(t) = (G_0 - G_\infty^N - \frac{\delta(Q_0 - Q_\infty^N)}{\sigma-\eta})e^{-\sigma t} + G_\infty^N + \frac{\delta(Q_0 - Q_\infty^N)}{\sigma-\eta}e^{-\eta t}$, *where* $Q_\infty^N = \frac{\alpha^2\delta\omega\pi_s}{\eta(\rho+\sigma)(\rho+\eta)k_f}$ *and* $G_\infty^N = \frac{\omega(\eta\beta^2(\eta+\rho)k_f\pi_r + \delta^2\alpha^2k_m\pi_s)}{\sigma\eta(\eta+\rho)(\rho+\sigma)k_fk_m}$ *are steady-state values of the product's freshness and goodwill.*

(iii) The steady-state profits for the supplier and the traditional platform are as follows: $V_{r\infty}^{N*} = m_1^*Q_\infty^N + m_2^*G_\infty^N + m_3^*$, $V_{s\infty}^{N*} = n_1^*Q_\infty^N + n_2^*G_\infty^N + n_3^*$, *where* $m_1^* = \frac{\delta\omega\pi_r}{(\sigma+\rho)(\eta+\rho)}$, $m_2^* = \frac{\omega\pi_r}{\sigma+\rho}$, $n_1^* = \frac{\delta\omega\pi_s}{(\sigma+\rho)(\eta+\rho)}$, $n_2^* = \frac{\omega\pi_s}{\sigma+\rho}$,
$m_3^* = \frac{\pi_r(2\alpha^2\delta^2\omega^2k_m\pi_s + \beta^2\omega^2k_f(\eta+\rho)^2\pi_r + 2Ak_fk_m(\eta+\rho)^2(\rho+\sigma)^2)}{2\rho k_fk_m(\eta+\rho)^2(\sigma+\rho)^2}$,
$n_3^* = \frac{\pi_s(2\beta^2\omega^2k_f(\eta+\rho)^2\pi_r + \alpha^2\delta^2\omega^2k_m\pi_s + 2Ak_fk_m(\eta+\rho)^2(\rho+\sigma)^2)}{2\rho k_fk_m(\eta+\rho)^2(\sigma+\rho)^2}$.

Proof. See Appendix A

Corollary 1. *The critical parameters' influence on the supplier's freshness-keeping effort and the platform's advertising effort is as follows:*

(i) $\frac{\partial\mu^{N*}}{\partial\xi} < 0$, $\frac{\partial\mu^{N*}}{\partial\delta} > 0$, $\frac{\partial\mu^{N*}}{\partial\omega} > 0$;

(ii) $\frac{\partial a^{N*}}{\partial\xi} < 0$, $\frac{\partial a^{N*}}{\partial\delta} < 0$, $\frac{\partial a^{N*}}{\partial\omega} > 0$.

Corollary 1 indicates that as ξ rises, consumers trust the platform more, leading to a decrease in the optimal freshness-keeping and advertising effort levels. With the influence of platform advertising, enhanced goodwill leads to higher consumer expectations. Additionally, the supplier may need to adjust its freshness efforts to align with the platform's advertising strategy. This coordination is essential for effectively operating the entire supply chain and maintaining product freshness.

In addition, as shown in Corollary 1, the optimal freshness-keeping increases with the increase of δ, but the advertising level decreases with the rise of δ. The increase in δ indicates that consumers focus more on freshness differences, reflecting a stronger RFE. In this case, both the supplier and platform adjust their strategies accordingly. Consequently, the supplier sets a higher freshness effort, while the platform reduces its advertising activities. This strategy enables RFE to positively impact the FPSC.

Lastly, Corollary 1 indicates that as ω increases, both freshness-keeping and advertising efforts are elevated. When consumers place a high value on goodwill, they often expect higher product freshness. This drives the supplier to invest more in freshness to meet these heightened expectations. Moreover, the platform may allocate more resources to advertising to enhance its brand image and meet consumer expectations for freshness.

Proof. See Appendix B

4.2. Strategic analysis for blockchain case (Model B)

Over an infinite horizon with a positive discount rate ρ, the supplier initially determines the freshness-keeping effort $\mu^B(t)$. The platform

$$m_4^* = \frac{(\delta(\omega-\theta\xi)+(\rho+\sigma)\theta)\pi_r}{(\rho+\sigma)(\eta+\rho)}, m_5^* = \frac{(\omega-\theta\xi)\pi_r}{\rho+\sigma}, n_4^* = \frac{(\delta(\omega-\theta\xi)+(\rho+\sigma)\theta)\pi_s}{(\rho+\sigma)(\eta+\rho)}, n_5^* = \frac{(\omega-\theta\xi)\pi_s}{\rho+\sigma},$$

$$m_6^* = \frac{(k_f(\eta+\rho)^2(\omega-\theta\xi)^2(\beta^2k_b+f^2k_m)\pi_r + 2\alpha^2k_mk_b(\delta(\omega-\theta\xi)+(\rho+\sigma)\theta)^2\pi_s + 2Ak_fk_mk_b(\eta+\rho)^2(\rho+\sigma)^2)\pi_r}{2k_fk_mk_b(\eta+\rho)^2(\sigma+\rho)^2},$$

$$n_6^* = \frac{(\alpha^2k_mk_b(\delta(\omega-\theta\xi)+\theta(\rho+\sigma))^2\pi_s + 2k_f(\eta+\rho)^2(\omega-\theta\xi)^2(\beta^2k_b+f^2k_m)\pi_r + 2Ak_fk_mk_b(\eta+\rho)^2(\rho+\sigma)^2)\pi_s}{2\rho k_fk_mk_b(\eta+\rho)^2(\sigma+\rho)^2}.$$

Box I.

subsequently determines the advertising effort $a^B(t)$ and the traceability level $b(t)$. Finally, consumers continuously place orders, and the quantities are denoted as $D^B(t)$. In this scenario, the objective functions of the supplier and the blockchain-based platform are given as follows:

$$V_s^B(t) = \int_0^\infty e^{-\rho t}\left\{\pi_s D^B(t) - \frac{k_f\mu^2(t)}{2}\right\}dt \tag{14}$$

$$V_r^B(t) = \int_0^\infty e^{-\rho t}\left\{\pi_r D^B(t) - \frac{k_m a^2(t)}{2} - \frac{k_b b^2(t)}{2}\right\}dt \tag{15}$$

Then, combining the dynamics of Eqs. (1)–(3), (6), (7), (9), (14) and (15), we obtain the following decision problems for the supplier and the blockchain-based platform under the model B.

$$\max_{\mu^B(t)} \int_0^\infty e^{-\rho t}\left\{\pi_s(A+\theta(Q(t)-r(t))+\omega G(t)) - \frac{k_f\mu^2(t)}{2}\right\}dt \tag{16}$$

$$\max_{a^B(t),b(t)} \int_0^\infty e^{-\rho t}\left\{\pi_r(A+\theta(Q(t)-r(t))+\omega G(t)) - \frac{k_m a^2(t)}{2} - \frac{k_b b^2(t)}{2}\right\}dt \tag{17}$$

To ensure feasibility, the condition $\omega-\theta\xi>0$ must hold. Using the Pontryagin maximization principle, we solve Eqs. (16)–(17) and derive Theorem 2.

Theorem 2. *In the blockchain-based FPSC, the following equilibrium solutions hold:*

(i) The optimal strategies for the supplier and the platform are as follows: $\mu^{B*} = \frac{\alpha((\omega-\theta\xi)\delta+(\rho+\sigma)\theta)\pi_s}{(\rho+\sigma)(\eta+\rho)k_f}$, $a^{B*} = \frac{\beta(\omega-\theta\xi)\pi_r}{(\rho+\sigma)k_m}$, $b^* = \frac{f(\omega-\theta\xi)\pi_r}{(\rho+\sigma)k_b}$.

(ii) The time trajectories of product freshness and blockchain-enabled goodwill are given by the following expressions: $Q^{B*}(t) = (Q_0-Q_\infty^B)e^{-\eta t} + Q_\infty^B$, $G^{B*}(t) = (G_0 - G_\infty^B - \frac{\delta(Q_0-Q_\infty^B)}{\sigma-\eta})e^{-\sigma t} + G_\infty^B + \frac{\delta(Q_0-Q_\infty^B)}{\sigma-\eta}e^{-\eta t}$, *where* $Q_\infty^B = \frac{\alpha^2((\omega-\theta\xi)\delta+(\rho+\sigma)\theta)\pi_s}{\eta(\eta+\rho)(\rho+\sigma)k_f}$ *and* $G_\infty^B = \frac{\alpha^2\delta k_mk_b((\omega-\theta\xi)\delta+(\rho+\sigma)\theta)\pi_s+\eta(\omega-\theta\xi)(\eta+\rho)(\beta^2k_b+f^2k_m)k_f\pi_r}{\sigma\eta(\eta+\rho)(\rho+\sigma)k_fk_mk_b}$ *are steady-state values of the product's freshness and goodwill.*

(iii) The steady-state profits of the supplier and the blockchain-based platform are given by the following expressions: $V_{r\infty}^{B*} = m_4^*Q_\infty^B + m_5^*G_\infty^B + m_6^*$, $V_{s\infty}^{B*} = n_4^*Q_\infty^B + n_5^*G_\infty^B + n_6^*$, *where* m_4^*, m_5^*, m_6^*, n_4^*, n_5^* *and* n_6^* *are given in Box* I

Proof. See Appendix A

Corollary 2. *The impact of key parameters on the supplier's optimal freshness-keeping effort, the platform's optimal advertising effort, and the platform's optimal traceability level is summarized as follows:*

(i) $\frac{\partial\mu^{B*}}{\partial\xi}<0$, $\frac{\partial\mu^{B*}}{\partial\delta}>0$, $\frac{\partial\mu^{B*}}{\partial\omega}>0$, $\frac{\partial\mu^{B*}}{\partial\theta}>0$;

(ii) $\frac{\partial a^{B*}}{\partial\xi}<0$, $\frac{\partial a^{B*}}{\partial\delta}<0$, $\frac{\partial a^{B*}}{\partial\omega}>0$, $\frac{\partial a^{B*}}{\partial\theta}<0$;

(iii) $\frac{\partial b^*}{\partial\xi}<0$, $\frac{\partial b^*}{\partial\delta}<0$, $\frac{\partial b^*}{\partial\omega}>0$, $\frac{\partial b^*}{\partial\theta}<0$.

The relationships between the supplier's optimal freshness-keeping efforts and the platform's optimal advertising effort concerning ξ, δ, and ω are similar to those under model N and will not be repeated here for brevity.

Meanwhile, as shown in Corollary 2(i) and (ii), the supplier's optimal freshness-keeping efforts are positively correlated with θ, while the platform's optimal advertising effort is negatively correlated with θ. When consumers' demand becomes more sensitive to product freshness, the supplier recognizes its importance to consumers. Under this condition, the platform may reduce its advertising efforts related to freshness and instead focus on other aspects of its brand or services to meet expectations.

Furthermore, the relationship between the platform's traceability level and the parameters ξ, δ, θ, and ω is derived. The traceability level is negatively correlated with ξ, implying that the platform may be less likely to adopt BT when customers rely more on goodwill to establish freshness expectations. In contrast, the traceability level increases with ω, indicating that when consumers highly value the platform's goodwill, it positively influences blockchain implementation. Customers who value the platform's goodwill may perceive BT as an additional layer of trust and transparency, thereby encouraging the platform to adopt it. However, if customers trust the platform's goodwill to ensure freshness, they may perceive additional transparency mechanisms, such as blockchain, as unnecessary.

Furthermore, the platform may reduce its willingness to invest in BT when RFE is pronounced (i.e., $\frac{\partial b^*}{\partial\delta}<0$, $\frac{\partial b^*}{\partial\theta}<0$*). Freshness-sensitive consumers place significant importance on the freshness of the products they purchase. The increase in RFE amplifies the discrepancy's impact between actual and advertised freshness, leading the platform to hesitate in disclosing additional information if it fears that BT will further highlight these trust issues.*

Proof. See Appendix B

5. Equilibrium comparison between two cases

This section identifies the necessary conditions for implementing BT on the platform and examines its impact on equilibrium strategies, freshness, brand goodwill, demand, and the supplier's profit in both scenarios. Building on the work of Ma et al. (2024), we define $\frac{f^2}{k_b}$, $\frac{\alpha^2}{k_f}$, and $\frac{\beta^2}{k_m}$ as the investment efficiencies of BT, freshness-keeping, and advertising, respectively. The main conclusions are summarized as follows.

5.1. Condition of BT implementation

Corollary 3. *The platform will adopt BT for product freshness disclosure if the following condition holds:*

$$\frac{f^2}{k_b} > \frac{F_1\alpha^2}{k_f} + \frac{F_2\beta^2}{k_m} + F_3.$$

Where

$$F_1 = \frac{2\pi_s\left(\begin{array}{l}\delta^2\eta\sigma(\rho-1)\omega^2 + ((\rho+\varphi)(\eta\sigma+\varphi) - \delta\xi(\eta-\varphi))\rho\theta^2(\rho+\varphi)\\ +\ \delta\rho\theta((\rho+\varphi)(2\sigma\eta+\eta+\rho+2\varphi) - \delta\xi(\eta-2\varphi-\rho))\omega\end{array}\right)}{\eta(\omega-\theta\xi)^2(\eta+\rho)^2(\sigma+2)\rho\pi_r},$$

$$F_2 = -\frac{\omega^2(\rho-1)\sigma - 2\xi\rho\theta(\sigma+2)\omega + \rho\theta^2\xi^2(\varphi+\xi+2)}{(\sigma+2)(\omega-\theta\xi)^2\rho},$$

$$F_3 = \frac{2(1-\rho)\sigma(\sigma+\rho)^2A}{\rho(\sigma+2)(\omega-\theta\xi)^2\pi_r}.$$

Corollary 3 defines the conditions under which the platform implements BT. When BT's investment efficiency is relatively large, the platform is strongly incentivized to adopt BT for higher profits. Higher BT's investment efficiency implies that the platform can attain better operational outcomes at an acceptable cost. As demonstrated in Corollary 2, under these conditions, the platform sets a higher traceability level. In contrast, when BT's investment efficiency falls below a certain threshold, the investment cost of BT outweighs its potential benefits, making the strategy economically unfeasible. This highlights the potential of BT in enhancing operational performance and fostering sustainable profitability.

This threshold is influenced not only by advertising and freshness-keeping investment efficiencies but also by the interaction of various factors, such as market demand, marginal benefits, and platform goodwill. Thus, when assessing the benefits of BT adoption, the platform must consider these interrelated variables, which may significantly affect the ultimate profitability and operational outcomes of BT implementation. Given these considerations, a careful evaluation of BT's investment efficiency is crucial for informed decision-making.

Proof. See Appendix B

5.2. Value of BT implementation

Corollary 4. *The impact of BT implementation on optimal freshness-keeping and advertising efforts is summarized as follows:* $\mu^{B*} > \mu^{N*}$ *and* $a^{B*} < a^{N*}$ *always hold.*

Corollary 4 indicates that the optimal freshness effort is higher with BT than without it. This occurs because BT implementation increases transparency and elevates quality expectations. Consequently, the supplier must enhance freshness management and allocate more resources to ensure product quality. This highlights the significant impact of BT on the supplier's investment decisions.

Similarly, the optimal advertising effort is lower with BT than without it. By adopting BT, the platform reduces its reliance on advertising and redirects resources to technology, achieving a more efficient and sustainable business model. This demonstrates blockchain's pivotal role in reducing information asymmetry and fostering consumer trust.

Corollary 5. *The impact of BT implementation on steady-state freshness and goodwill is as follows:*

(i) $Q_{\infty}^{B*} > Q_{\infty}^{N*}$ *is always holds.*

(ii) $G_{\infty}^{B*} > G_{\infty}^{N*}$, *if* $\frac{f^2}{k_b} > F_4$; *otherwise,* $G_{\infty}^{B*} < G_{\infty}^{N*}$. *Where* $F_4 = -\frac{\theta\pi_s\delta(\rho+\varphi)\alpha^2}{\eta\pi_r(\omega-\theta\xi)(\eta+\rho)k_f} + \frac{\xi\theta\beta^2}{(\omega-\theta\xi)k_m}$.

Corollary 5 asserts that the FPSC achieves higher equilibrium freshness in blockchain-supported environments compared to traditional settings. As highlighted in Corollary 4, BT encourages the supplier to improve its freshness-keeping efforts through real-time monitoring of product conditions and environmental influences. These improved steady-state values play a significant role in advancing freshness-keeping strategies.

Moreover, Corollary 5 emphasizes that sufficiently high investment efficiency in implementing BT increases goodwill. In contrast, low investment efficiency results in reduced goodwill. This phenomenon may arise from the "double effect" of blockchain, which simultaneously fosters goodwill through transparency while reducing reliance on advertising. Higher BT's investment efficiency enhances product freshness initiatives, generating substantial goodwill by strengthening consumer trust. Furthermore, the transparency inherent in BT can partially replace the trust-building functions traditionally fulfilled by advertising. This alleviates potential adverse effects arising from reduced advertising expenditures.

Conversely, when investment efficiency is low, consumers respond less to BT-provided information. Limited advertising further exacerbates trust deficits, as consumers lack alternative reliable information sources, ultimately reducing goodwill.

Proof. See Appendix B

Corollary 6. *The impact of BT implementation on steady-state demand is described as follows:* $d_{\infty}^{B*} > d_{\infty}^{N*}$, *if* $\frac{f^2}{k_b} > F_5$; *otherwise,* $d_{\infty}^{B*} < d_{\infty}^{N*}$. *Where* $F_5 = -\frac{\theta\pi_s(\delta\omega\rho + 2\delta\omega\varphi + \rho\theta\varphi + \theta\varphi^2)\alpha^2}{\eta\pi_r(\omega-\theta\xi)^2(\eta+\rho)k_f} + \frac{(2\omega-\theta\xi)\xi\theta\beta^2}{(\omega-\theta\xi)^2k_m}$.

Corollary 6 highlights the effect of BT's investment efficiency on consumer demand. Without BT, consumers primarily rely on goodwill to form their expectations of freshness, which are influenced by the platform's historical reputation and market standing. In this scenario, quality uncertainty persists.

By introducing BT, consumers can access verifiable product information, reducing freshness uncertainty. However, the effectiveness of BT depends on its investment efficiency. When BT's investment efficiency exceeds a critical threshold, the platform can provide consumers with authentic and reliable product information. This transparency strengthens consumer trust, enhances goodwill, and elevates demand above the level observed without BT. However, if BT's investment efficiency falls below the threshold, consumer demand will decline due to insufficient reliability of blockchain data.

Proof. See Appendix B

Corollary 7. *The impact of BT implementation on steady-state supplier profits is described as follows:* $V_{s\infty}^{B*} > V_{s\infty}^{N*}$, *if* $\frac{f^2}{k_b} > F_6$; *otherwise,* $V_{s\infty}^{B*} < V_{s\infty}^{N*}$. *Where*

$$F_6 = -\frac{\theta\pi_s\alpha^2\begin{pmatrix}2\delta(\eta\varphi\sigma + \rho(\rho+2\varphi)(\sigma+\eta+\rho))\omega \\ -\theta(\rho+\varphi)(\xi\eta\delta\rho - \varphi(\sigma+\rho)(\eta+2\rho))\end{pmatrix}}{2\eta\pi_r(\omega-\theta\xi)^2(\eta+\rho)^2(\sigma+\rho)k_f} + \frac{(2\omega-\theta\xi)\xi\theta\beta^2}{(\omega-\theta\xi)^2k_m}.$$

Corollary 7 shows that the supplier's long-term profits can only be enhanced by BT implementation when BT's investment efficiency surpasses a certain threshold. This threshold is affected by the efficiency of advertising investment and the efficiency of freshness-keeping activities. Specifically, the higher the efficiency of advertising investment, the lower the threshold for the supplier to benefit, and the higher the efficiency of freshness-keeping activities, the higher the threshold for the supplier to benefit. When the effectiveness of freshness-keeping efforts is significant, the costs incurred by the supplier to maintain freshness are already quite low, making the addition of BT potentially offer minimal added value. Consequently, in a scenario where the efficiency of freshness-keeping efforts is high, the threshold for investment efficiency in BT must also be elevated to ensure that the supplier can obtain its benefits.

Proof. See Appendix B

6. Numerical analysis

The results discussed in Section 4 are complex and multifaceted. This makes direct comparisons challenging. Therefore, this section employs numerical analysis to provide a clearer comparison and draw comprehensive conclusions.

The parameter settings in this section are based on established values from previous related studies (Li et al., 2023b; Zhao et al., 2023), ensuring consistency and reliability in the analysis:

$A = 1$, $\alpha = 0.5$, $\beta = 0.3$, $\rho = 0.2$, $\eta = 0.2$, $\varphi = 0.2$, $\xi = 0.5$, $\omega = 0.5$, $\theta = 0.6$, $\delta = 0.3$, $f = 0.5$, $k_f = 2$, $k_m = 2$, $k_b = 2$.

6.1. Time trajectories of freshness and goodwill

To explore the impact of initial freshness and goodwill on equilibrium, we consider the following four combinations of initial conditions: $Q_0 = 1 < Q_{\infty}^j$ and $G_0 = 1 < G_{\infty}^j$, $Q_0 = 7 > Q_{\infty}^j$, and $G_0 = 1 < G_{\infty}^j$, $Q_0 = 1 < Q_{\infty}^j$, and $G_0 = 7 > G_{\infty}^j$, and $Q_0 = 7 > Q_{\infty}^j$, and $G_0 = 7 > G_{\infty}^j$ respectively, where superscript $j \in \{N, B\}$ and time

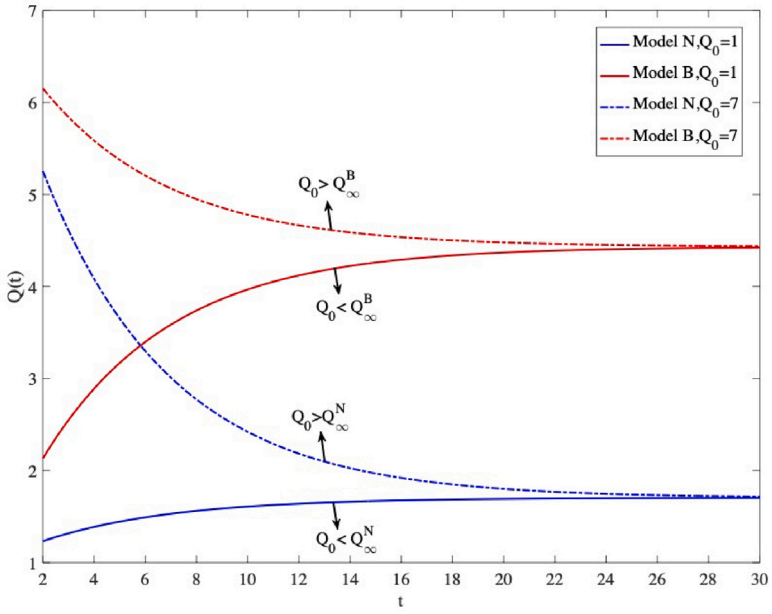


Fig. 2. Product freshness trajectories with different initial freshness.

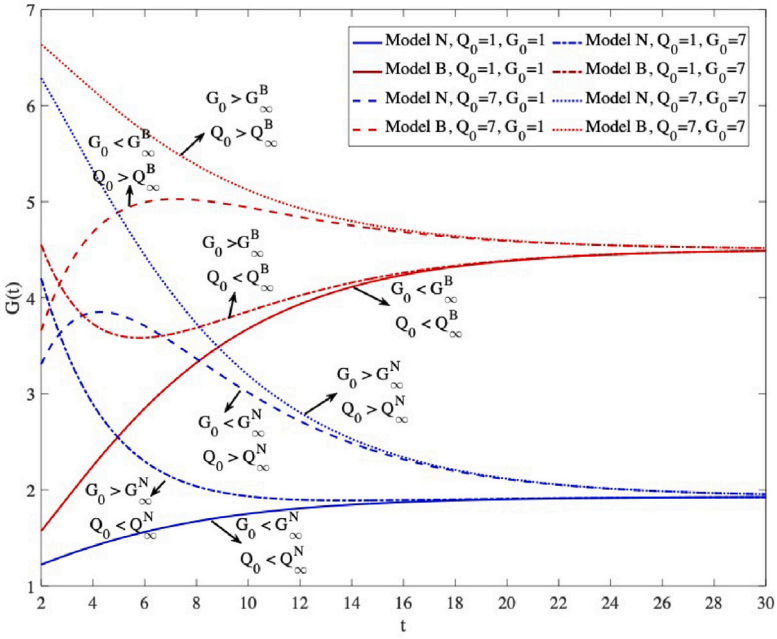
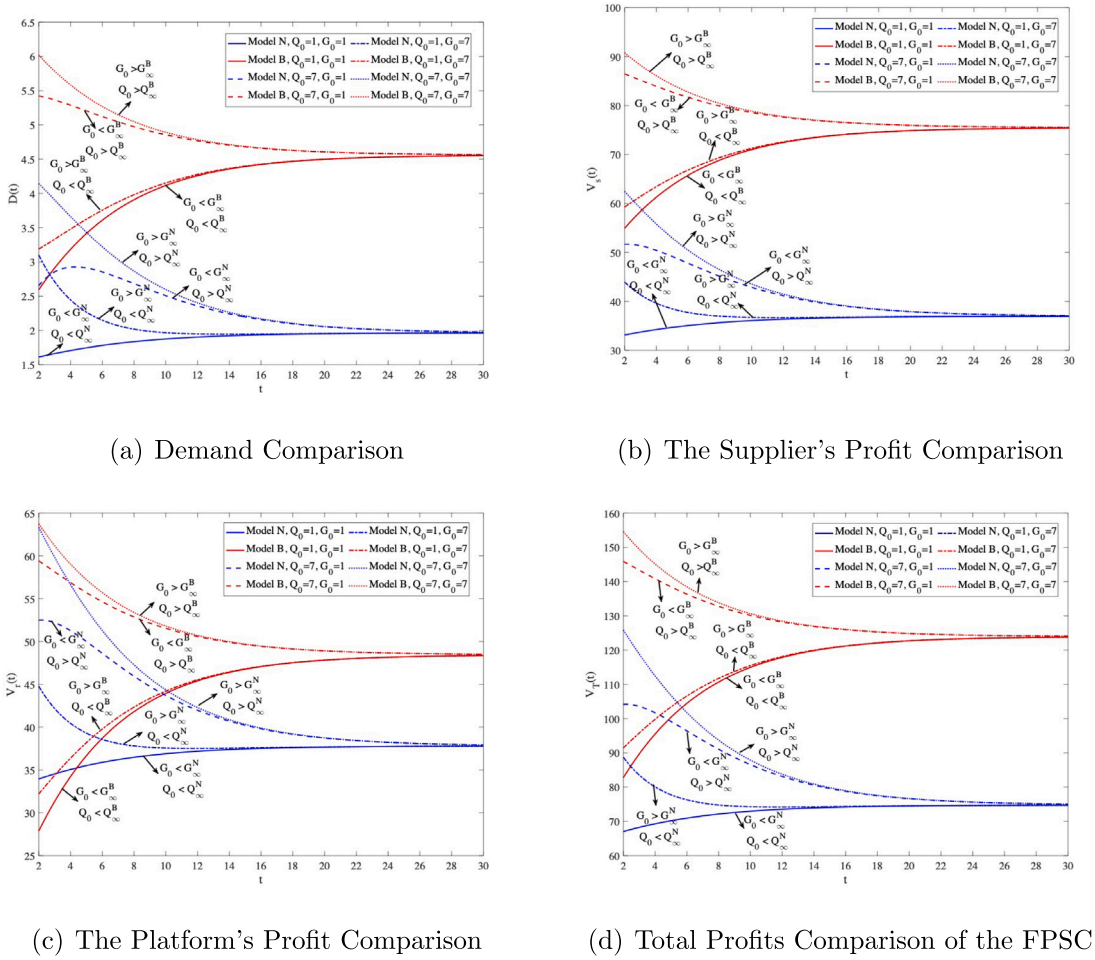


Fig. 3. Goodwill trajectories with different initial freshness and goodwill.

$t \in [2, 30]$. We analyze the impact of different initial conditions of freshness and goodwill on the equilibrium state of the system and its long-term stability.

Figs. 2 and 3 show that as time t approaches infinity, the optimal product freshness $Q^{j*}(t)$ and goodwill $G^{j*}(t)$ converge to a steady state. Regardless of the initial freshness Q_0, the steady-state value of $Q^{j*}(t)$ always remains consistent. Notably, $Q^{B*}(t) > Q^{N*}(t)$ always holds. This reemphasizes that BT helps products achieve higher levels of freshness. Moreover, Fig. 2 shows that the change in $Q^{j*}(t)$ is monotonically varying, as proved by Theorems 1 and 2. Specifically, when $Q_0 < Q^j_\infty$, the product freshness increases and reaches a steady state. Conversely, if initial freshness $Q_0 > Q^j_\infty$, the product freshness decreases to an equilibrium state. This observation highlights the direct relationship between the initial freshness level and the final steady-state value. Fig. 3 shows the trajectory of goodwill under two scenarios. When both initial freshness and initial goodwill are higher than their respective steady-state levels ($Q_0 > Q^j_\infty, G_0 > G^j_\infty$), goodwill shows a monotonically decreasing trend. In this case, BT can slow down the decay rate, allowing the system to maintain a high level of goodwill for a longer period of time. When initial freshness is higher than the steady-state level but initial goodwill is lower than the steady-state level ($Q_0 > Q^j_\infty, G_0 < G^j_\infty$), goodwill first decreases and then increases. This shows that BT accelerates the recovery process, allowing goodwill to reach its steady-state level faster and more efficiently.

When initial freshness is lower than the steady-state level but initial goodwill exceeds the steady-state level ($Q_0 < Q^j_\infty, G_0 > G^j_\infty$), goodwill initially increases and prolongs the time that goodwill remains above the steady-state level, thus stabilizing system performance. Finally, when both initial freshness and initial goodwill are below their respective steady-state levels ($Q_0 < Q^j_\infty, G_0 < G^j_\infty$), goodwill increases



(a) Demand Comparison

(b) The Supplier's Profit Comparison

(c) The Platform's Profit Comparison

(d) Total Profits Comparison of the FPSC

Fig. 4. The comparison of demand and FPSC profits under different scenarios.

monotonically. In this case, BT accelerates the recovery process and promotes a faster recovery of trust, allowing the system to reach a steady-state goodwill level more quickly.

From Fig. 3 we can conclude that the BT application improves the final goodwill level of the system and accelerates the convergence process. This positive effect is particularly evident under adverse conditions ($Q_0 < Q_\infty^j, G_0 < G_\infty^j$). These findings highlight the potential of blockchain in enhancing trustworthiness and promoting long-term stability and operational efficiency in FPSC dynamic systems.

6.2. Integral analysis of the efficiency of the FPSC

Fig. 4 compares the demand, member profit, and total profit of traditional and blockchain models. Fig. 4 (a) illustrates that demand is susceptible to changes in freshness and goodwill. Demand rapidly decreases when $Q_0 < Q_\infty^j$ and $G_0 < G_\infty^j$, while it rapidly increases when $Q_0 > Q_\infty^j$ and $G_0 > G_\infty^j$. This behavior arises from the direct influence of consumers' sensitivity to freshness and goodwill on demand. Blockchain demonstrates a significant optimization effect on demand fluctuations by not only reducing the rate of decline and accelerating recovery but also improving the final steady-state level of demand.

Fig. 4 (b), (c), and (d) depict the dynamic changes in supplier profit, platform profit, and total profit, respectively. The supplier profit change curve shows a smoother decline or rise compared to demand. The platform profit change curve is similar to the supplier's but exhibits lower volatility. As a comprehensive supply chain performance indicator, total profit demonstrates the most significant blockchain optimization effect.

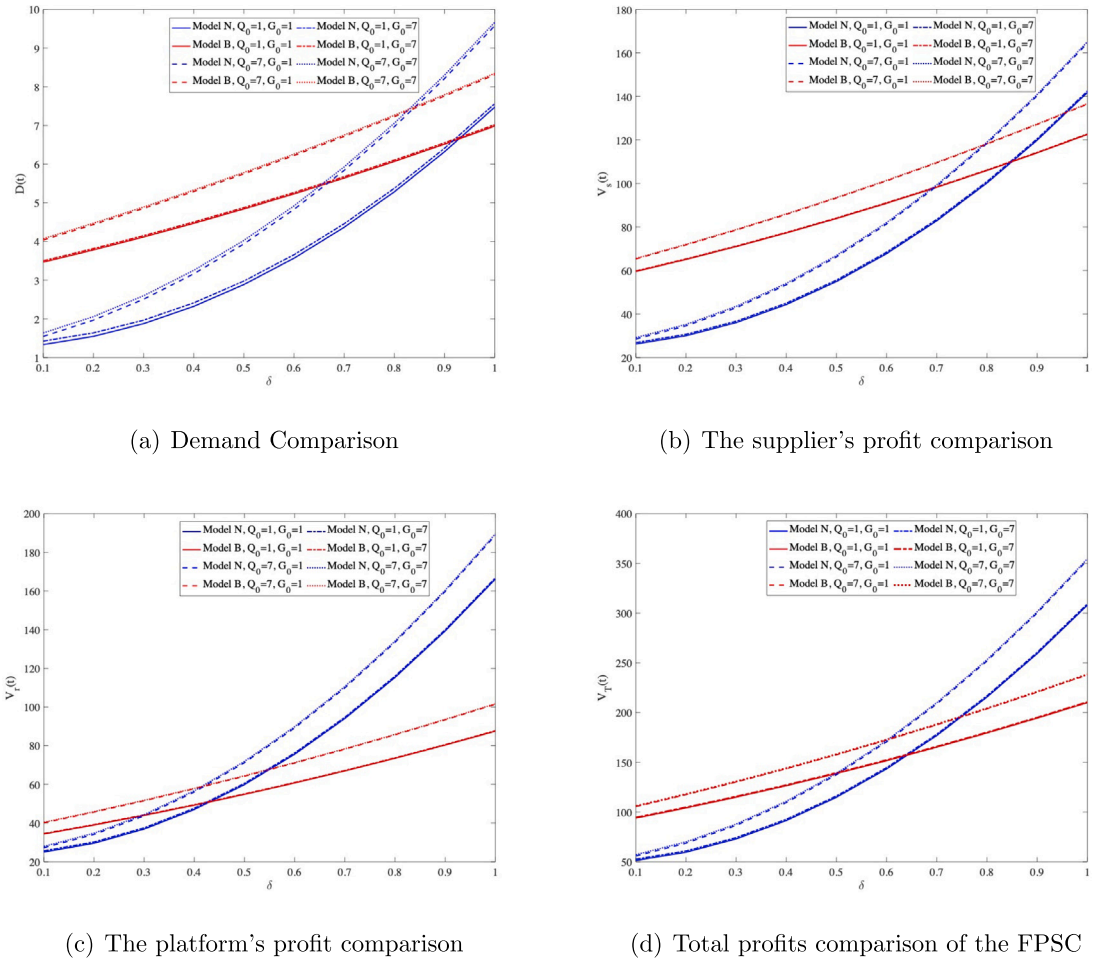
6.3. Impact of RFE on the demand and profits

For sensitivity analysis, we selected $t = 10$ and $\delta \in (0, 1]$ as the experimental case. Fig. 5 outlines the impact of RFE on demand, member profit, and total profit under blockchain and non-blockchain scenarios. Fig. 5 shows that demand and profit increase with the rise of RFE. BT moderates this impact by enhancing transparency and narrowing the actual-perceived quality gap. Additionally, the influence of RFE on demand and profit is less sensitive to initial freshness because it is primarily driven by initial goodwill. Adopting BT yields more favorable results when initial goodwill is high, as it enhances trust and reduces information asymmetry.

Furthermore, Fig. 5 (b), (c), and (d) illustrate that FPSC members will choose to adopt BT if RFE is sufficiently small. In this case, the overall profit performance of the supply chain improves. In contrast, when BT is not adopted, the impact of RFE on members and overall profits is more significant, creating a clear distinction compared to the scenario of adopting BT. This difference demonstrates the role of BT in alleviating the adverse effects of RFE.

6.4. Sensitivity analysis of marginal benefits

For sensitivity analysis, we selected $t = 10, [\pi_s, \pi_r] \in [2, 6]$ as the experimental case. Fig. 6 summarizes the impact of marginal benefits on demand, member profits, and total profits. In Fig. 6 (a), demand is relatively insensitive to increases in π_s and π_r in the traditional model. However, with the implementation of BT, demand becomes extremely sensitive to π_s. This suggests that BT reduces information asymmetry



(a) Demand Comparison

(b) The supplier's profit comparison

(c) The platform's profit comparison

(d) Total profits comparison of the FPSC

Fig. 5. The impact of RFE on the demand and profits.

and enhances supplier efficiency, leading to greater demand sensitivity to π_s.

Fig. 6 (b) and (c) illustrate the impact of marginal benefits on FPSC members in both blockchain-based and traditional models. The supplier's profits show higher sensitivity to π_s, particularly in the blockchain model, as BT improves the efficiency of all participants in the supply chain, especially at the supplier level. However, platform profits are more affected by π_r.

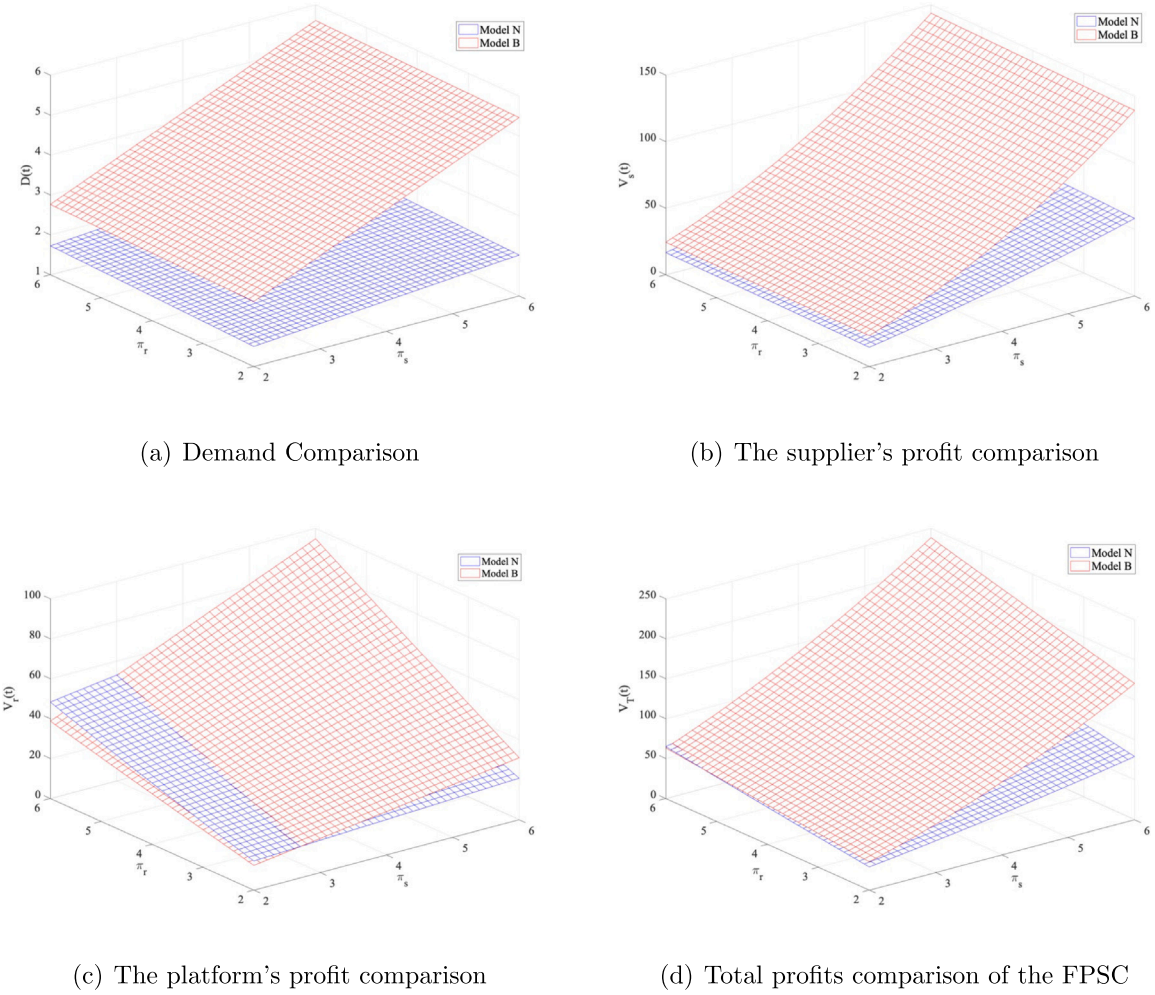
Moreover, Fig. 6 (d) highlights that total FPSC profit is more sensitive to π_s than π_r, emphasizing the supplier's critical role in determining profitability. The blockchain-based model demonstrates a faster growth rate, reflecting its superior ability to adapt to market changes and achieve higher long-term profitability.

7. Conclusion

Inspired by emerging practices in the fresh e-commerce industry, this paper proposes a dynamic decision management model for a fresh product supplier and an e-commerce platform in the presence of freshness reference effects among consumers. In model N, the fresh product supplier and the e-commerce platform decide on freshness and advertising efforts, while consumers make purchase decisions based on uncertain freshness information. Businesses can adjust their information traceability levels and decisions using BT in model B. Subsequently, consumers make purchase decisions based on updated freshness information.

Our findings reveal that when customers heavily rely on the platform's goodwill to shape their expectations, optimal freshness-keeping and advertising levels decrease. As RFE increases, the optimal freshness-keeping effort level increases, while advertising effort decreases, and the platform may be less inclined to adopt a higher level of BT traceability. Furthermore, optimal freshness-keeping efforts are elevated with BT, demonstrating its significant impact on the supplier's freshness-keeping investment decisions. Simultaneously, advertising efforts tend to decline with BT implementation, reducing advertising costs significantly. Under steady-state conditions, the freshness achieved surpasses the traditional case when BT is implemented. When the BT implementation efficiency is sufficiently high, it results in greater goodwill. When considering the numerical example, we additionally observe that BT enables the supply chain to adjust and adapt to market changes more quickly.

The following are critical managerial insights. First, when the supplier joins a system that adopts BT, they should also allocate additional resources to maintain freshness. For instance, MOA Planet, a South Korean O2O blockchain-based e-commerce platform, enables its suppliers with higher product freshness to disclose their freshness levels, distinguishing them from regular suppliers. Additionally, platforms can utilize BT to provide verifiable information, thereby reducing advertising activities. Meanwhile, managers are required to evaluate the balance between goodwill, freshness expectations, and BT's investment efficiency to inform business decisions. For example, during a leafy green safety crisis, Walmart deployed BT to reduce the traceability time for contaminated products from days to seconds. This rapid response alleviated consumer concerns and reinforced Walmart's goodwill as a reliable platform. Moreover, collaboration between suppliers and platforms is essential to optimizing strategies, ensuring product quality, and meeting customer needs. The impact of RFE



(a) Demand Comparison

(b) The supplier's profit comparison

(c) The platform's profit comparison

(d) Total profits comparison of the FPSC

Fig. 6. The impact of π_s and π_r on the demand and profits.

depends on joint decisions made by suppliers and platforms, such as optimizing freshness-keeping measures, advertising strategies, and BT implementation. Management can be improved by promoting collaboration, transparency, and freshness. Additionally, FPSC operational investment efficiency can be optimized, ensuring continued success in the ever-changing market landscape.

Combining the freshness reference effects with blockchain will produce meaningful and exciting outcomes. Future research can focus on the following directions. First, a model can be established from the perspective of BT to evaluate additional quality attributes of fresh food, such as sweetness and taste, and assess blockchain's effectiveness in improving consumer satisfaction. Second, the coordination contract between supply chain members should be explored. This includes focusing on effectively combining the freshness reference effect with BT through coordination and cooperation, further optimizing supply chain management and product quality control.

CRediT authorship contribution statement

Xiujuan Liu: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation. **Yong He:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Golnaz Hooshmand Pakdel:** Writing – review & editing. **Shanshan Li:** Writing – review & editing, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Proof of Theorem 1

In scenario N, we aim to show that the value functions V_s^N and V_r^N are continuously differentiable. The Hamilton–Jacobi–Bellman (HJB) equations for the supplier and platform can be expressed as:

$$\rho V_s^N = \max_{\mu^N} \left\{ \pi_s D^N - \frac{k_f \mu^2}{2} + \frac{\partial V_s^N}{\partial Q}(\alpha\mu - \eta Q) + \frac{\partial V_r^N}{\partial G}(\beta a + \delta Q - \sigma G) \right\}, \tag{A.1}$$

$$\rho V_r^N = \max_{a^N} \left\{ \pi_r D^N - \frac{k_m a^2}{2} + \frac{\partial V_s^N}{\partial Q}(\alpha\mu - \eta Q) + \frac{\partial V_r^N}{\partial G}(\beta a + \delta Q - \sigma G) \right\}. \tag{A.2}$$

By applying the first-order optimality condition to Eq. (A.2), the optimal decision for the platform's advertising effort is:

$$a^N = \frac{\beta \frac{\partial V_r^N}{\partial G}}{k_m}. \tag{A.3}$$

By substituting this into Eq. (A.1), the supplier's freshness-keeping effort is obtained as:

$$\mu^N = \frac{\alpha \frac{\partial V_s^N}{\partial Q}}{k_f}. \tag{A.4}$$

Substitute Eqs. (A.3) and (A.4) into the HJB equations of the supplier and the platform, respectively, to obtain the following set of HJB equations:

$$\rho V_s^N(Q,G) = (\delta \frac{\partial V_s^N}{\partial G} - \eta \frac{\partial V_s^N}{\partial Q})Q - (\sigma \frac{\partial V_s^N}{\partial G} - \omega \pi_r)G + \pi_r A + \frac{\beta^2 \frac{\partial V_s^N}{\partial G}^2}{2k_m} + \alpha\mu \frac{\partial V_s^N}{\partial Q}, \tag{A.5}$$

$$\rho V_r^N(Q,G) = (\delta \frac{\partial V_r^N}{\partial G} - \eta \frac{\partial V_r^N}{\partial Q})Q + (\pi_s \omega - \sigma \frac{\partial V_r^N}{\partial G})G + \pi_s A + \frac{\alpha^2 \frac{\partial V_s^N}{\partial Q}^2}{2k_f} + \frac{\beta^2 \omega \pi_r \frac{\partial V_s^N}{\partial G}^2}{(\sigma+\rho)k_m}. \tag{A.6}$$

By observing the structure of Eqs. (A.5) and (A.6), we can see that $V_r^N(Q,G)$ and $V_s^N(Q,G)$ are linear functions of the first order in Q and G. Since the HJB equations yield linear combinations of Q and G, the linear expressions of $V_r^N(Q,G)$ and $V_s^N(Q,G)$ are:

$$\rho V_s^N(Q,G) = n_1 Q + n_2 G + n_3, \tag{A.7}$$

$$\rho V_r^N(Q,G) = m_1 Q + m_2 G + m_3, \tag{A.8}$$

where n_1, n_2, n_3, m_1, m_2, m_3, are unknown constants. Substituting Eqs. (A.7) and (A.8) into Eqs. (A.5) and (A.6), and using the method of coefficients to be determined, we get:

$$\begin{aligned}
m_1^* &= \frac{\delta\omega\pi_r}{(\sigma+\rho)(\eta+\rho)}, \quad m_2^* = \frac{\omega\pi_r}{\sigma+\rho}, \quad n_1^* = \frac{\delta\omega\pi_s}{(\sigma+\rho)(\eta+\rho)}, \quad n_2^* = \frac{\omega\pi_s}{\sigma+\rho}, \\
m_3^* &= \frac{\pi_r\left(2\alpha^2\delta^2\omega^2 k_m \pi_s + \beta^2\omega^2 k_f(\eta+\rho)^2\pi_r + 2Ak_f k_m(\eta+\rho)^2(\rho+\sigma)^2\right)}{2\rho k_f k_m(\eta+\rho)^2(\sigma+\rho)^2}, \\
n_3^* &= \frac{\pi_s\left(2\beta^2\omega^2 k_f(\eta+\rho)^2\pi_r + \alpha^2\delta^2\omega^2 k_m\pi_s + 2Ak_f k_m(\eta+\rho)^2(\rho+\sigma)^2\right)}{2\rho k_f k_m(\eta+\rho)^2(\sigma+\rho)^2}.
\end{aligned} \tag{A.9}$$

Substituting Eq. (A.9) into Eqs. (A.3) and (A.4), the optimal strategies for the supplier and the platform are obtained as:

$$\mu^{N*} = \frac{\alpha\delta\omega\pi_s}{(\rho+\sigma)(\eta+\rho)k_f}, a^{N*} = \frac{\beta\omega\pi_r}{(\sigma+\rho)k_m}. \tag{A.10}$$

Substituting Eq. (A.10) into Eq. (1), the optimal trajectory of freshness under model N is

$$Q^{N*}(t) = (Q_0 - Q_\infty^N)e^{-\eta t} + Q_\infty^N, \tag{A.11}$$

where $Q_\infty^N = \frac{\alpha^2\delta\omega\pi_s}{\eta(\rho+\sigma)(\rho+\eta)k_f}$ represents the steady-state of the product's freshness.

Substituting Eqs. (A.10) and (A.11) into Eq. (5), the optimal trajectory of goodwill under model N is described as:

$$G^{N*}(t) = (G_0 - G_\infty^N - \frac{\delta(Q_0 - Q_\infty^N)}{\sigma-\eta})e^{-\sigma t} + G_\infty^N + \frac{\delta(Q_0 - Q_\infty^N)}{\sigma-\eta}e^{-\eta t}, \tag{A.12}$$

where $G_\infty^N = \frac{\omega(\eta\beta^2(\eta+\rho)k_f\pi_r + \delta^2\alpha^2 k_m\pi_s)}{\sigma\eta(\eta+\rho)(\rho+\sigma)k_f k_m}$ denotes the steady-state brand goodwill.

The steady-state profits for the supplier and the traditional platform, as described in Theorem 1, are derived accordingly.

Similarly, Theorem 2 can be proved using the same method.

Appendix B

B.1. Proof of Corollary 1

Based on the results in Theorem 1, we can verify the relationships: $\frac{\partial \mu^{N*}}{\partial \xi} = \frac{-\alpha\omega\delta^2\pi_s}{(\sigma+\rho)^2(\eta+\rho)k_f} < 0$, $\frac{\partial \mu^{N*}}{\partial \delta} = \frac{\alpha\omega(\rho+\varphi)\pi_s}{(\sigma+\rho)^2(\eta+\rho)k_f} > 0$, $\frac{\partial \mu^{N*}}{\partial \omega} = \frac{\alpha\delta\pi_s}{(\sigma+\rho)(\eta+\rho)k_f} > 0$;
Likewise,we can obtain $\frac{\partial a^{N*}}{\partial \xi} < 0$, $\frac{\partial a^{N*}}{\partial \delta} < 0$, $\frac{\partial a^{N*}}{\partial \omega} > 0$.
Corollary 2 can be proved in the same way.

B.2. Proof of Corollary 3

The equilibrium solution obtained from Theorem 1 gives the difference in state-steady platform profits between the traditional and blockchain-based conditions $V_{r\infty}^{B*} - V_{r\infty}^{N*} = f^2 - E_1 k_b > 0$,where $E_1 = \frac{F_1\alpha^2}{k_f} + \frac{F_2\beta^2}{k_m} + F_3$,

$$F_1 = \frac{2\pi_s\left(\begin{array}{l}\delta^2\eta\sigma(\rho-1)\omega^2 + ((\rho+\varphi)(\eta\sigma+\varphi) - \delta\xi(\eta-\varphi))\rho\theta^2(\rho+\varphi) \\ + \delta\rho\theta\big((\rho+\varphi)(2\sigma\eta+\eta+\rho+2\varphi) - \delta\xi(\eta-2\varphi-\rho)\big)\omega\end{array}\right)}{\eta(\omega-\theta\xi)^2(\eta+\rho)^2(\sigma+2)\rho\pi_r},$$

$$F_2 = -\frac{\omega^2(\rho-1)\sigma - 2\xi\rho\theta(\sigma+2)\omega + \rho\theta^2\xi^2(\varphi+\xi+2)}{(\sigma+2)(\omega-\theta\xi)^2\rho},$$

$$F_3 = \frac{2(1-\rho)\sigma(\sigma+\rho)^2 A}{\rho(\sigma+2)(\omega-\theta\xi)^2\pi_r}$$

Corollaries 4–7 can be proved in the same way.

Data availability

No data was used for the research described in the article.

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